

problem 4.4: Formulating an MDP

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Grid Reference

Using (x, y) coordinates with $(1, 1)$ at bottom-left:

$(1, 3)$	$(2, 3)$	$(3, 3)$	$(4, 3)$	← row 3 (top)
$(1, 2)$	$(2, 2)$	$(3, 2)$	$(4, 2)$	← row 2 (middle)
$(1, 1)$	$(2, 1)$	$(3, 1)$	$(4, 1)$	← row 1 (bottom)

- $(4, 3)$: goal, reward $+1$, terminal
- $(4, 2)$: hazard, reward -1 , terminal
- $(2, 2)$: wall (not a valid state)
- All other cells: reward -0.04
- Start: $(1, 1)$

Part 1: State and Action Spaces

States: All grid cells except the wall at $(2, 2)$:

$$\mathcal{S} = \{(1, 1), (2, 1), (3, 1), (4, 1), (1, 2), (3, 2), (4, 2), (1, 3), (2, 3), (3, 3), (4, 3)\}$$

That is 11 states total.

Terminal states: $(4, 3)$ (goal) and $(4, 2)$ (hazard).

Non-terminal states: $11 - 2 = 9$.

Actions: $\mathcal{A} = \{\text{North, South, East, West}\}$.

Part 2: Transition Probabilities from (1, 2) Taking East

From (1, 2), the intended action is East (toward (2, 2)).

- **Intended (East, $p = 0.8$):** (2, 2) is a wall $\rightarrow$ stay at (1, 2).
- **Drift left/North ($p = 0.1$):** (1, 3) is a valid cell $\rightarrow$ move to (1, 3).
- **Drift right/South ($p = 0.1$):** (1, 1) is a valid cell $\rightarrow$ move to (1, 1).

Since the intended move is blocked, that probability stays at the current cell:

$$\begin{aligned} T((1, 2) \mid (1, 2), \text{East}) &= 0.8 && \text{(intended blocked by wall)} \\ T((1, 3) \mid (1, 2), \text{East}) &= 0.1 && \text{(drift north)} \\ T((1, 1) \mid (1, 2), \text{East}) &= 0.1 && \text{(drift south)} \end{aligned}$$

Check: $0.8 + 0.1 + 0.1 = 1.0\checkmark$

The wall at (2, 2) makes East from (1, 2) essentially a "stay" action with small chances of drifting north or south.

Part 3: Bellman Optimality Equation for (3, 2)

$$V^*((3, 2)) = R((3, 2)) + \gamma \max_a \sum_{s'} T(s' \mid (3, 2), a) V^*(s')$$

With $R((3, 2)) = -0.04$, we expand two actions:

Action North (intended: (3, 3), drift left: (2, 2) wall, drift right: (4, 2)):

- Intended North $\rightarrow$ (3, 3): probability 0.8
- Drift West $\rightarrow$ (2, 2) is a wall $\rightarrow$ stay at (3, 2): probability 0.1
- Drift East $\rightarrow$ (4, 2): probability 0.1

$$\sum_{s'} T(s' \mid (3, 2), \text{N}) V^*(s') = 0.8 V^*((3, 3)) + 0.1 V^*((3, 2)) + 0.1 V^*((4, 2))$$

Note that (4, 2) is the hazard with $V^*((4, 2)) = -1$. This means going North from (3, 2) has a 10% chance of entering the hazard.

Action East (intended: (4, 2), drift left: (3, 3), drift right: (3, 1)):

- Intended East $\rightarrow$ (4, 2): probability 0.8
- Drift North $\rightarrow$ (3, 3): probability 0.1
- Drift South $\rightarrow$ (3, 1): probability 0.1

$$\sum_{s'} T(s' | (3, 2), E) V^*(s') = 0.8 V^*((4, 2)) + 0.1 V^*((3, 3)) + 0.1 V^*((3, 1))$$

This sends the robot into the hazard with 80% probability—clearly a poor choice.

The full Bellman equation selects the max over all four actions:

$$V^*((3, 2)) = -0.04 + \gamma \max(\text{N-value}, \text{S-value}, \text{E-value}, \text{W-value})$$

Part 4: Value Iteration by Hand

Use $\gamma = 1$.

Initialization: $V_0(s) = 0$ for non-terminal states, $V_0((4, 3)) = +1$, $V_0((4, 2)) = -1$.

Iteration 1

For every non-terminal state s :

$$V_1(s) = R(s) + \gamma \max_a \sum_{s'} T(s' | s, a) V_0(s')$$

Since $V_0 = 0$ for all non-terminal states, only transitions into terminal states contribute to the sum. A state can only get a nonzero expected value if one of its actions can reach $(4, 3)$ or $(4, 2)$.

States adjacent to $(4, 3)$: $(3, 3)$ and $(4, 3)$ itself (terminal, skip). From $(3, 3)$, East reaches $(4, 3)$ with $p = 0.8$.

States adjacent to $(4, 2)$: $(3, 2)$, $(4, 1)$, and $(4, 3)$ (terminal, skip). From $(3, 2)$, East reaches $(4, 2)$ with $p = 0.8$. From $(4, 1)$, North reaches $(4, 2)$ with $p = 0.8$.

Let's compute V_1 for each non-terminal state by finding the best action:

$(3, 3)$: Best action is East. - East:

$$0.8 \times V_0((4, 3)) + 0.1 \times V_0((3, 3)) + 0.1 \times V_0((3, 2)) = 0.8 \times 1 + 0 + 0 = 0.8 -$$

North: hits wall with 0.8, drift to $(2, 3)$ with 0.1, drift to $(4, 3)$ with 0.1

$$= 0 + 0 + 0.1 \times 1 = 0.1 - V_1((3, 3)) = -0.04 + 1 \times 0.8 = 0.76$$

$(2, 3)$: Best action is East. - East:

$$0.8 \times V_0((3, 3)) + 0.1 \times V_0((2, 3)) + 0.1 \times V_0((2, 2)).$$

But $(2, 2)$ is a wall, so drift south stays at $(2, 3)$. All V_0 values are 0. - No action reaches a terminal state, so

$$\max_a(\dots) = 0. - V_1((2, 3)) = -0.04 + 0 = -0.04$$

$(1, 3)$: No action reaches a terminal state. - $V_1((1, 3)) = -0.04$

(3, 2): Best action needs care—East goes directly to the hazard.

- East: intended (4, 2), drift-N (3, 3), drift-S (3, 1):
 $0.8 \times (-1) + 0.1 \times 0 + 0.1 \times 0 = -0.8$
- North: intended (3, 3), drift-W (2, 2) wall $\rightarrow$ stay (3, 2), drift-E (4, 2):
 $0.8 \times 0 + 0.1 \times 0 + 0.1 \times (-1) = -0.1$
- South: intended (3, 1), drift-W (2, 2) wall $\rightarrow$ stay (3, 2), drift-E (4, 2):
 $0.8 \times 0 + 0.1 \times 0 + 0.1 \times (-1) = -0.1$
- West: intended (2, 2) wall $\rightarrow$ stay (3, 2), drift-N (3, 3), drift-S (3, 1):
 $0.8 \times 0 + 0.1 \times 0 + 0.1 \times 0 = 0$

Best is West with value 0. $V_1((3, 2)) = -0.04 + 0 = -0.04$.

(4, 1): Adjacent to the hazard—several actions risk drifting into (4, 2).

- North: intended (4, 2) hazard, drift-W (3, 1), drift-E off-grid $\rightarrow$ stay:
 $0.8 \times (-1) + 0.1 \times 0 + 0.1 \times 0 = -0.8$
- East: intended off-grid $\rightarrow$ stay, drift-N (4, 2) hazard, drift-S off-grid $\rightarrow$ stay:
 $0.8 \times 0 + 0.1 \times (-1) + 0.1 \times 0 = -0.1$
- West: intended (3, 1), drift-N (4, 2) hazard, drift-S off-grid $\rightarrow$ stay:
 $0.8 \times 0 + 0.1 \times (-1) + 0.1 \times 0 = -0.1$
- South: intended off-grid $\rightarrow$ stay, drift-W (3, 1), drift-E off-grid $\rightarrow$ stay:
 $0.8 \times 0 + 0.1 \times 0 + 0.1 \times 0 = 0$

Best is South with value 0. (South is safe because all outcomes avoid the hazard.)

$V_1((4, 1)) = -0.04$.

Remaining states $\{(1, 1), (2, 1), (3, 1), (1, 2)\}$: No action from any of these can reach a terminal state in one step (they are too far away). So $\max_a(\dots) = 0$ and $V_1 = -0.04$ for all of them.

Summary of V_1 :

State	V_1
(4, 3)	+1.00 (terminal)
(4, 2)	−1.00 (terminal)
(3, 3)	+0.76
All others	−0.04

Iteration 2 for (3, 2)

$$V_2((3, 2)) = -0.04 + \max_a \sum_{s'} T(s' | (3, 2), a) V_1(s')$$

North: intended (3, 3), drift-W stays (wall at (2, 2)), drift-E → (4, 2):

$$0.8 \times 0.76 + 0.1 \times (-0.04) + 0.1 \times (-1) = 0.608 - 0.004 - 0.1 = 0.504$$

South: intended (3, 1), drift-W stays (wall), drift-E → (4, 2):

$$0.8 \times (-0.04) + 0.1 \times (-0.04) + 0.1 \times (-1) = -0.032 - 0.004 - 0.1 = -0.136$$

East: intended (4, 2), drift-N → (3, 3), drift-S → (3, 1):

$$0.8 \times (-1) + 0.1 \times 0.76 + 0.1 \times (-0.04) = -0.8 + 0.076 - 0.004 = -0.728$$

West: intended stays (wall at (2, 2)), drift-N → (3, 3), drift-S → (3, 1):

$$0.8 \times (-0.04) + 0.1 \times 0.76 + 0.1 \times (-0.04) = -0.032 + 0.076 - 0.004 = 0.040$$

Best action: **North** with value 0.504.

$$V_2((3, 2)) = -0.04 + 0.504 = 0.464$$

Iteration 2 for (3, 1)

$$V_2((3, 1)) = -0.04 + \max_a \sum_{s'} T(s' | (3, 1), a) V_1(s')$$

North: intended (3, 2), drift-W → (2, 1), drift-E → (4, 1):

$$0.8 \times (-0.04) + 0.1 \times (-0.04) + 0.1 \times (-0.04) = -0.04$$

South: hits wall, stay. Drift-W → (2, 1), drift-E → (4, 1):

$$0.8 \times (-0.04) + 0.1 \times (-0.04) + 0.1 \times (-0.04) = -0.04$$

East: intended (4, 1), drift-N → (3, 2), drift-S hits wall, stay:

$$0.8 \times (-0.04) + 0.1 \times (-0.04) + 0.1 \times (-0.04) = -0.04$$

West: intended (2, 1), drift-N → (3, 2), drift-S hits wall, stay:

$$0.8 \times (-0.04) + 0.1 \times (-0.04) + 0.1 \times (-0.04) = -0.04$$

All actions are tied at -0.04 because (3, 1) is surrounded by states that all have $V_1 = -0.04$. The high value at (3, 3) hasn't propagated this far yet.

$$V_2((3, 1)) = -0.04 + (-0.04) = -0.08$$

Part 5: Qualitative Policy Sketch

+-----+	+-----+	+-----+	+-----+
→	→	→	GOAL
+-----+	+-----+	+-----+	+-----+
↑	WALL	↑	HAZD
+-----+	+-----+	+-----+	+-----+
↑	→	↑	←
+-----+	+-----+	+-----+	+-----+

Reasoning:

- **Top row ($y = 3$):** Move East toward the goal. These states are safe (no risk of drifting into the hazard from the top row).
- **(1, 2):** Move North to reach the safe top row. The wall at **(2, 2)** blocks eastward progress, so the robot must go around.
- **(3, 2):** This is the most dangerous state. East goes directly to the hazard. North goes toward **(3, 3)**, which is near the goal, but drifting East from North leads to the hazard with 10% probability. The optimal policy likely chooses North (accepting the 10% hazard risk) because the alternative paths are much longer.
- **(4, 1):** North goes to the hazard with 80% probability. The robot should move **West** to retreat from the hazard column before heading north.
- **(3, 1):** The robot should go North toward **(3, 2)**, which despite its proximity to the hazard, is the most direct route.

Where the policy differs from the shortest path: At **(4, 1)**, the shortest path to the goal is North then North-then-East, but the stochastic transitions make this extremely dangerous (80% chance of hitting the hazard on the first step). The optimal policy retreats West first. This is the same safety-vs-efficiency tradeoff we see in the **4 × 4** warehouse MDP.

Most cautious states: **(3, 2)** and **(4, 1)**, both of which are adjacent to the hazard at **(4, 2)**.

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